

Understanding on the creep behavior of fiber reinforced polymer via fiber/matrix interaction

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ABSTRACT

The interfacial interaction between fiber and matrix is believed as one of the key parameters that affects viscoelastic response of different types of fiber reinforced polymer (FRP) materials under constant loads, which origins from molecular level. Current experimental observations mostly attribute the differing viscoelastic responses of various FRPs to stress redistribution between fibers, while the differences in creep behaviors among various types of FRPs have not been well understood. Key factors in fiber/matrix interfacial interactions, such as the mobility of polymer chains, microstructural evolution at the interface, and the energy changes during the creep process in different FRP systems remain unclear. To comprehensively understand the interfacial creep behavior between fiber and matrix, this work investigates carbon fiber/epoxy and glass fiber/epoxy composites using molecular dynamics simulations. The study aims to elucidate the deterioration mechanism of the fiber/matrix interface under constant loads and highlight the differences in creep performance between carbon FRP and glass FRP. Results reveal that carbon FRP demonstrates a higher shear threshold stress under equivalent holding load conditions. Energy analysis and bond autocorrelation functions indicate higher non-bonded interactions and lower the creep migration rate of epoxy at carbon fiber/matrix interface. These findings offer insights into the molecular-level origins of interfacial creep behavior in composites, providing potential solutions to enhance the viscoelastic properties of FRP composites.

1. Introduction

The application of fiber-reinforced polymer (FRP) composites in civil engineering to reinforce concrete structures, both externally and internally, has drawn attention from researchers and industry professionals. This interest stems from the outstanding properties of FRP, notably its exceptional stiffness-to-weight ratio and excellent corrosion resistance [1,2]. Among various applications, FRP rebars are widely used as alternatives to traditional steel reinforcement, particularly in challenging service environments [3]. However, uncertainty remains about the long-term performance of FRP composites under constant loads due to the varying creep behaviors reported in the literature for different types of FRP composites. This uncertainty has led to overly conservative designs for FRP reinforcement in civil infrastructure [4,5]. The potential

implications of creep in FRP composites raise concerns about structural integrity [6]. With the increasing usage of FRP composites in construction industry, it is critical to thoroughly understand the long-term behavior of FRP under sustained service loads, along with developing strategies to improve their resistance to creep deformation.

The creep response of FRP composites generally progresses through three distinct phases: initial, intermediate, and advanced stages. The initial and intermediate stages are critical as they significantly affect the creep deformation and service life of composite structures. It has been reported that the initial creep is influenced by matrix cracking, while subsequent creep is affected by fiber-matrix detachment. Fibers play a crucial role during the final creep stage, which is mainly characterized by fiber fracture [7,8]. However, even with the same polymer matrix material, such as epoxy resin, significant differences in creep properties

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are reported between carbon FRP (CFRP) and glass FRP (GFRP) composites under the same constant load levels, as documented in various studies [9–12]. During the initial 30-day creep stage, CFRP rebars exhibit lower creep ratios than GFRP rebars, highlighting the superior performance of CFRP rebars under constant loads. This conclusion is drawn from tensile tests conducted on CFRP and GFRP rebars with an epoxy matrix under equivalent load conditions [12]. These findings suggest that the interaction at the fiber/matrix interface significantly influences the first two stages, primarily through interfacial creep phenomena, which are challenging to capture using conventional experimental methods.

To understand the interfacial failure of CFRP under the constant loads, efforts have focused on fiber pull-out test. Studies have observed that at lower applied forces, fiber displacement increases gradually during the pulling process. As the applied force increases, the rate of fiber displacement accelerates, highlighting interfacial debonding as a major failure mechanism [13–15]. In addition to macroscopic creep tests conducted on bulk composite samples, investigations of local interfacial creep are performed using nano-indentation techniques. These results indicate that when applied loads exceed a critical threshold, cracking occurs at the fiber/matrix interface. Leading to significant displacement increases and fiber slippage, a phenomenon commonly referred to as creep rupture. This stage is characterized by a critical stress level known as interfacial shear strength [16–21]. These experiments demonstrate that interfacial shear strength plays a crucial role in the creep process. When the applied load surpasses a specific threshold, the interface experiences slip damage, and microstructural observations confirm the fiber push-outs.

As aforementioned in the micromechanical tests, the creep at the fiber/matrix interface and the detachment are considered critical factors in determining different creep behaviors of various types of FRP composites [22,23]. The results from micromechanical tests, such as fiber pull-out or nanoindentation tests, only suggest that interfacial creep may influence both the initial and intermediate stages. These experiments primarily attribute this effect to stress redistribution between fibers. However, they do not provide a clear understanding of the specific mechanisms driving creep rupture or how the interfacial strength threshold impacts the creep process. Furthermore, they do not fully explain how the interfacial creep is affected by the interaction between polymer matrix and different fiber surfaces [24,25]. Therefore, comprehensively understanding the variation in mechanical properties and microstructural changes at the fiber/matrix interface under constant loads is crucial for explaining the mechanisms that contribute to the observed differences in creep behavior among various FRP composites. This understanding is closely linked to the molecular-level interactions between polymer chains and atoms exposed on the surface of fiber substrate at the nanoscale.

Molecular dynamics (MD) simulation is a fundamental method to study molecular interactions in FRP composites, providing valuable insights into interfacial behavior and the mechanisms of debonding [26–31]. Tam et al. [32,33] utilized MD simulations to investigate the interfacial microstructure and deformation behavior of CFRP composites. Jian et al. [34] developed a model to explore the interface deformation process in GFRP composites using an epoxy/silica interface configuration. The investigation focused on studying the extension and sliding behavior at epoxy/silica interface under varying loads. The results suggest that the contact area between the epoxy matrix and the silica substrate significantly influences interface behavior, spanning from the initiation of creep deformation to eventual debonding failure. Wang et al. [35] utilized an epoxy/glass interface model consisting of epoxy resin with amorphous silica. Under humid and thermal conditions, their findings indicate that epoxy molecules near the fiber surface experience reduced density due to softening. This reduction in stress transmission from the fiber to the matrix ultimately weakens adhesion at the interface. Although recent research has provided insights on the variation of interfacial properties in FRP under environmental or

mechanical loads, the mechanisms underlying the differences in creep behavior between various types of FRP composites are still not well understood.

This study aims to elucidate the mechanism behind the different interfacial creep behaviors of FRP composites at nanoscale, using CFRP and GFRP as examples. Specifically, interface models are developed to represent CFRP and GFRP composites, with a focus on the carbon fiber/matrix and glass fiber/matrix interfaces. Subsequently, shearing creep tests are then conducted using MD simulations. Through the analysis using an energy-based approach and local segmental dynamics, this study characterized the creep behavior of carbon fiber/epoxy and glass fiber/epoxy composites. The characterization includes time-dependent stress-strain behavior, and the adhesion energy associated with both bonded and non-bonded interactions, along with microstructural analysis. The simulation results reveal that the carbon fiber/epoxy interface demonstrates a larger threshold force and higher adhesion energy compared to the glass fiber/matrix interface, underscoring its superior resistance to creep. These findings offer insights into the molecular-level origins of interfacial creep behavior in composites, offering potential solutions to enhance the viscoelastic properties of FRP composites.

2. Model and simulation details

2.1. Molecular model

To investigate the creep behavior of carbon fiber/matrix and glass fiber/matrix composites, two interface models are constructed at the nanoscale. The graphite carbon fiber surface is represented by graphite sheets, the glass fiber surface is represented by a silica substrate, and the epoxy is modeled using cross-linked molecules [36–38], as depicted in Fig. 1(a). The epoxy molecular model is established using diglycidyl ether of bisphenol A (DGEBA) and triethylenetetramine (TETA) as the essential monomers for forming the epoxy cross-linked network. When DGEBA and TETA are mixed, covalent bonds can form between carbon (C) and nitrogen (N) atoms when their distance ranges from 3 Å to 10 Å. This occurs due to the generation of reactive CH₂ sites created by breaking C–O bonds in the epoxide group of DGEBA; the reaction atoms of DGEBA and TETA are labeled in this cross-linking procedure, as shown in Fig. 1(b). Fig. 1(c) presents the chemical and atomistic structures. The resulting cross-linked epoxy molecule measures 4.9 × 5.9 × 5.9 nm, consists of a total of 16926 atoms, which is sufficiently large for interfacial simulations [34]. The modeling is performed using Material Studio (MS). The perl script is employed to implement the cross-linking process in MS for constructing a densely cross-linked epoxy network. The degree of cross-linking reaches 85.3 %, which agrees with the experimentally observed cross-linking degree of about 80 % [38].

For modeling carbon fiber, non-periodic graphite sheets consisting of five layers are constructed. Each layer measures 7.8 × 7.7 nm and is capped with hydrogen atoms on its sides. The graphite model has a thickness of 1.4 nm, exceeding the 1.0 nm cutoff distance for non-bonded interactions. It ensures the model accurately represents the behavior of carbon fiber [39–42], particularly at the interface where the majority of the carbon fiber's bulk has minimal influence. For the glass fiber, an amorphous silica substrate is modeled, as depicted in Fig. 1(a). The structural information about the silica model refers to neutron scattering experiments [41]. The initial crystalline silicon dioxide with dimensions 3.0 × 6.5 × 6.5 nm, undergoes a melting process for over 3 ns, with the temperature linearly increasing from 298 K to 4998 K in the isothermal and isobaric (NPT) ensemble. Following this, it is equilibrated for 1 ns at 4998 K to ensure complete elimination of the original crystal structure. Subsequently, it is slowly cooled down to 298 K using a quench rate of 5 × 10¹² K/s in the NPT ensemble, promoting rapid solidification. This specific quench rate is selected to yield a reliable structural description for amorphous silica [36]. The silica model is then cut into a substrate with a size similar to that of the graphite model, and the exposed unsaturated sites are terminated by hydrogen atoms to

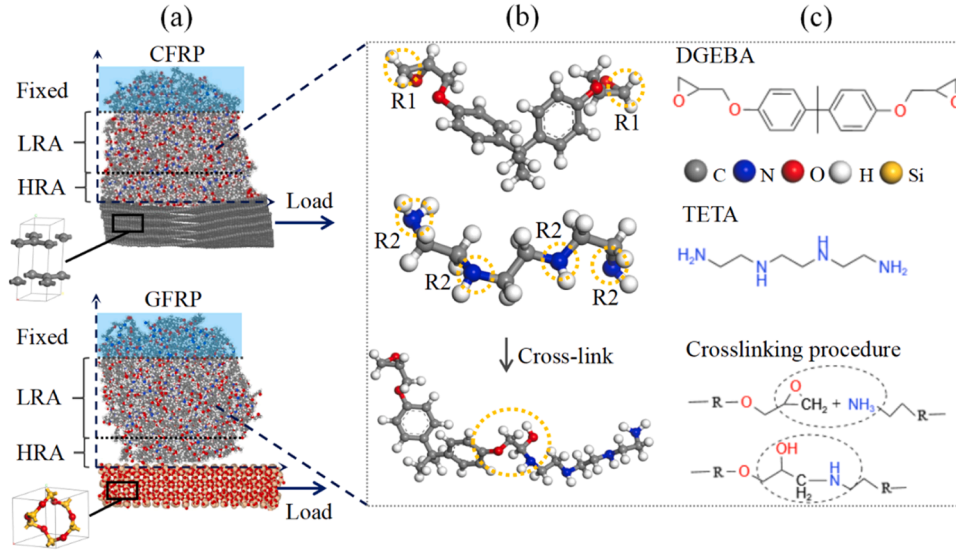


Fig. 1. (a) Bilayer system with 25 % of the upper segment of the epoxy matrix being fixed. The HRA is located within 10 Å from the lower surface of the fiber substrate exhibits the strongest bonding, primarily through van der Waals and Coulombic interactions. Moreover, the region between the HRA and the fixed area is termed the LRA; (b) the reaction atoms of DGEBA and TETA. The bond between the active oxygen atom R1 and carbon atom is cleaved, while forming a new carbon-nitrogen bond with the nitrogen atom R2 in TETA simultaneously; and (c) chemical structure of DGEBA and TETA and cross-linking process.

remove periodicity.

With the components ready, the carbon and glass fiber/matrix interface models are assembled, following methodologies similar to those used in previous studies on interfacial sliding and debonding [43, 44]. These models are placed at the center of a $30 \times 30 \times 30$ nm box, ensuring that mirror models do not affect the creep simulations [28,32, 33].

2.2. Force field

In this study, the Consistent Valence Force Field (CVFF) is used to calculate atomic-level interactions within this system. The CVFF includes a comprehensive set of energy terms to describe bonded interactions, covering bonds, angles, dihedrals, as well as improper terms. Additionally, non-bonded interactions are governed by van der Waals (vdW) and Coulombic terms. The CVFF potential function is simplified by adopting the harmonic form, as described in Eq. 1 [44]:

$$E = \sum_b K_b (b - b_0)^2 + \sum_\theta K_\theta (\theta - \theta_0)^2 + \sum_\phi K_\phi (1 + \text{sco} \sin \phi) + \sum_{ij} \epsilon_{ij} \left[\left(\frac{r_{ij}^*}{r_{ij}} \right)^{12} - \left(\frac{r_{ij}^*}{r_{ij}} \right)^6 \right] + \sum_{ij} \frac{q_i q_j}{\epsilon r_{ij}} \quad (1)$$

where K_b, K_θ, K_ϕ represent force constants; b, b_0 represent bond length; θ, θ_0, ϕ represent bond angle; ϵ_{ij} denotes the depth of the vdW potential well; r_{ij}^*, r_{ij} denotes the length when the vdW potential reaches its minimum state and distance between particle i and particle j with charges q_i and q_j . During the simulation, vdW and short-range Coulombic interactions are cutoff at 10 Å, where the interaction energy between non-bonded atoms is minimal. Long-range Coulombic interactions are computed using a particle-particle particle-mesh (PPPM) solver [45], while the partial charges of atoms are determined using the bond increment method [46]. The widespread application of CVFF in modeling and simulating epoxy-bonded systems has demonstrated significant agreement with experimental findings, confirming its accuracy and practicality [47,48].

2.3. Equilibration and validation

The software LAMMPS is used for equilibration, with a timestep

configured to 1 fs [48]. Firstly, the molecular model undergoes relaxation in the microcanonical (NVE) ensemble for 1 ns, with the fiber substrate layer held fixed. Subsequently, it relaxes in the canonical (NVT) ensemble, where the temperature rises from 0 K to 573 K and then cools back to 298 K. Throughout the NVT equilibration phase, the temperature is adjusted in 80 K increments, with the simulation running for 0.5 ns at each increment. After the NVT equilibration, the model undergoes NPT equilibration in 298 K for 1 ns. Finally, the fiber substrate layer is released and undergoes the same equilibrium process sequentially in NVT and NPT ensembles. Temperature control is achieved using the Nosé-Hoover thermostat, while pressure is controlled using the Barostat [32,49,50]. During equilibration, the densities of epoxy molecule and amorphous silica stabilize at 1.046 g/cm³ and 2.20 g/cm³, respectively, aligning with macroscopic experimental results [50].

In this study, the glass transition temperature (T_g) is utilized to validate the epoxy molecule, while radial distribution functions (RDF) are calculated for the amorphous silica. The epoxy model has the initial density of 1.135 g/cm³. To examine the T_g of the epoxy model, a gradual cooling process is performed from 643 K to 250 K, decreasing in increments of 25 K. The results are gathered to build the density-temperature curve at each recorded temperature following a 200 ps equilibration period. The significant shift in the density-temperature curve for the epoxy model reveals a T_g of 436 K, closely matching experimental values of 440 K [39,51,52], as shown in Fig. 2(a). Thus, the model is validated for subsequent simulation studies. Meanwhile, the RDF for the amorphous silica molecule model is calculated, as shown in Fig. 2(b). The reduction in the first peak value indicates that the simulated silica substrate has achieved an amorphous state.

2.4. Creep simulation setup

The steered molecular dynamics (SMD) simulations are employed to investigate the interfacial creep behavior of carbon fiber/matrix and glass fiber/matrix interface following the equilibration process. SMD is an advanced simulation technique that mimics the atomic force microscope (AFM) to study interactions between materials, particularly in bonded systems. SMD is widely recognized as an effective simulation method for investigating creep behavior [32–34]. During the SMD simulation, a pulling force is applied to the center-of-mass (COM) of the

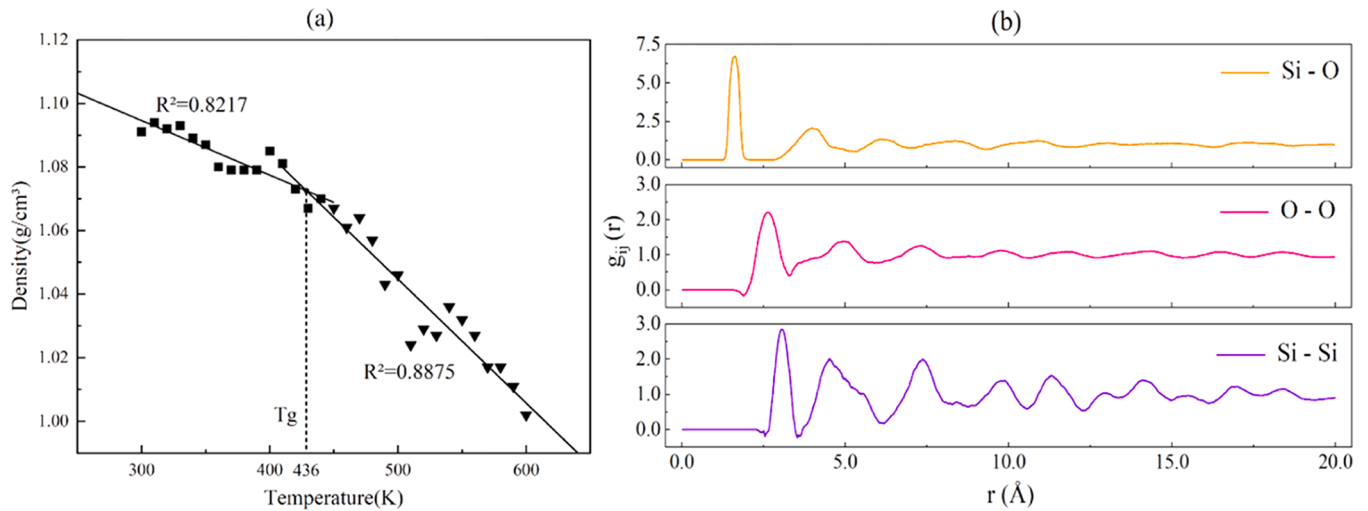


Fig. 2. (a) The glass transition temperature of the cross-linked epoxy; and (b) partial radial distribution functions of amorphous silica.

fiber substrate to simulate shearing, while the top 25 % layer of the epoxy model thickness is held fixed. To minimize atomic oscillations and ensure the fiber substrate responds appropriately, a harmonic force with a spring constant of 200.0 kcal/mol/Å² is applied to stabilize the substrate at its original positions [33,43], as shown in Fig. 1(a). Throughout the entire creep simulation at each force level, the displacement of the fiber substrate, changes in microstructure, and the system's energy are monitored and recorded. The simulation concludes either upon detachment or after reaching 30 ns, chosen to ensure a stable deformed state. Extended simulations up to 100 ns validate that 30 ns is sufficient for accurate creep analysis.

3. Results and discussion

In this section, the creep behaviors of carbon fiber/matrix and glass fiber/matrix interfaces are measured and compared through shear creep testing. Simultaneously, a strength-based approach is employed to characterize the threshold stress for interfacial detachment and the microscopic structural evolution between carbon fiber/matrix and glass fiber/matrix interface. Subsequently, an energy-based method is used to characterize the creep response processes. The local segmental dynamics of polymer chains in the epoxy matrix are quantified using autocorrelation functions, revealing the creep mechanism differences between the carbon fiber/matrix and glass fiber/matrix interface.

3.1. Threshold stress of interfacial creep

In the creep simulation, the fiber substrate is subjected to a constant force applied instantaneously at various load levels. The simulation at each load level continues until either the fiber substrate completely detaches, or the simulation reaches 30 ns, at which point a stable and constant displacement condition is achieved. This stable state is regarded as adequately extended compared to the multi-nanosecond time-scales used in prior creep simulations of different materials [32–34]. During this simulation, the peak displacement of the fiber substrate is determined by measuring the maximum distance between the COM of the fiber substrate and its initial position. Fig. 3 illustrates the displacement in relation to the applied force graphically. Significantly, the dataset is classified into two regimes characterized by complete detachment observed for both the graphite and silica substrates. Under the low load regime, below 1600 pN, the displacement of the graphite substrate increases linearly with the applied force, while the interface remains intact. Conversely, under the high load regime, observed with loads exceeding 1700 pN, the displacement of the graphite substrate eventually exceeds the dimensions of the epoxy molecule, leading to complete detachment between graphite substrate and epoxy matrix. Similarly, for the silica substrate, critical thresholds of 250 pN and 300 pN define the low and high load regimes, respectively.

Under shear loading, a critical force threshold exists before interfa-

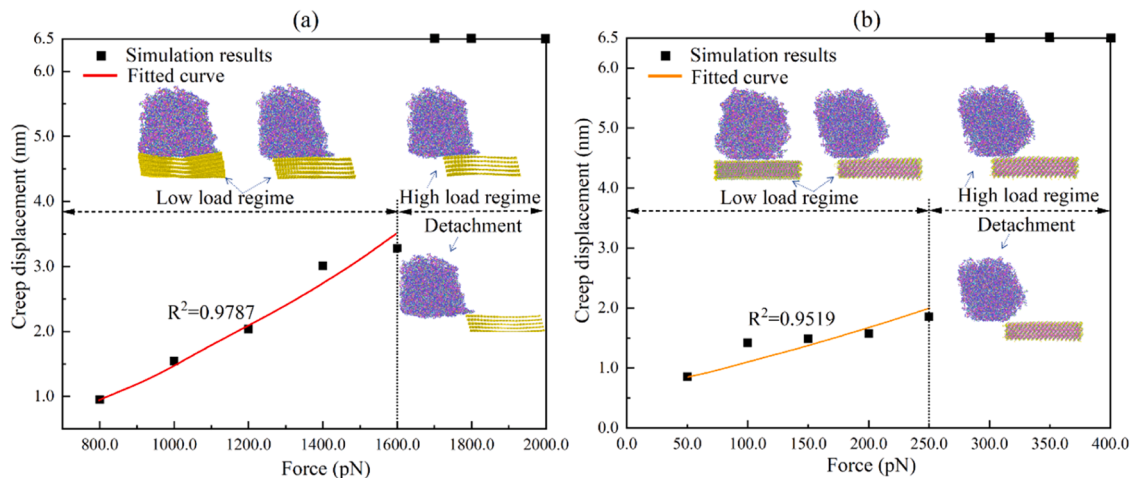


Fig. 3. The creep displacement plotted against the applied load levels for (a) CFRP and (b) GFRP.

cial detachment is achieved: 1600–1700 pN for the carbon fiber/matrix interface and 250–300 pN for the glass fiber/matrix interface. The threshold stress causing interfacial detachment is determined by calculating the engineering stress with equation $S = F/A$, here F represents the applied force on the fiber substrate, A represents the contact area between the fiber substrate and epoxy matrix. Importantly, the threshold stress represents the maximum stress the interface can withstand without detachment under constant loads, reflecting its resistance to interfacial creep. The results indicate that the shearing threshold stress for the carbon fiber/matrix interface falls between 32.95 MPa and 35.01 MPa (with $A = 45.56 \text{ nm}^2$) and for the glass fiber/matrix interface, it ranges from 17.04 MPa to 20.45 MPa (with $A = 14.67 \text{ nm}^2$). Remarkably, both shearing threshold stresses obtained from the simulation agree well with experimentally tested interfacial shear strengths. These experimental values, determined from pull-out or nano-indentation tests of CFRP and GFRP rods, yielded values of $39.3 \pm 5.4 \text{ MPa}$ and $15.73 \pm 6.84 \text{ MPa}$, respectively [19,20,53–55]. The slight difference could be attributed to structural voids in actual material samples, resulting in weaker interfacial properties observed experimentally [56–58]. Emphasizing the importance of threshold stress as the maximum load-bearing capacity of the interface without detachment under constant loads, both simulation and experimental results demonstrate a higher threshold stress for the carbon fiber/epoxy interface compared to the glass fiber/epoxy interface. This observation highlights the superior shearing threshold stress in the carbon fiber/epoxy interface, playing a significant role in explaining the observed differences in creep performance between carbon fiber/epoxy and glass fiber/epoxy composites.

To gain deeper insights and validate the creep behavior over an extended duration, simulations were extended to 100 ns. It was found

that the creep displacement remained relatively constant, approximately 3.40 nm at 32.95 MPa and 1.90 nm at 17.04 MPa, while the interface remains intact. This stability indicates that the creep displacement reaches a constant value throughout the extended simulation period. Therefore, these findings suggest that the initial 30 ns duration is adequate to examine the atomistic-level creep behavior.

3.2. Comparison of interfacial creep evolution

To comprehensively illustrate the distinct interfacial creep evolution processes in carbon fiber/epoxy and glass fiber/epoxy composites, the interfacial creep response is characterized by examining microstructural changes and strain variations during creep under both low and high load regimes. In particular, the interfacial strain under shear loading is determined by normalizing the displacement of the fiber substrate with respect to the dimensions of the equilibrated interfacial model in both vertical and lateral directions. Fig. 4 presents the strain-time curves and microstructural changes observed in carbon fiber/epoxy and glass fiber/epoxy interface obtained from SMD simulations conducted under constant load conditions. Under the low load regime, the interfacial creep response of the carbon fiber/epoxy and glass fiber/epoxy interface is depicted at representative force levels, ranging from 800 pN to 1600 pN and 50–250 pN, respectively. Similarly, under the high load regime, the force levels vary from 1700 to 2000 pN for carbon fiber/epoxy interface and from 300 pN to 400 pN for glass fiber/epoxy interface. Furthermore, our focus is directed towards contrasting strain variations at force levels of 1600 pN and 1700 pN for the carbon fiber/epoxy interface, and at force levels of 250 pN and 300 pN for the glass fiber/epoxy interface, corresponding to observed stress thresholds. Atoms in the epoxy molecules are color-coded according to their individual strain levels to

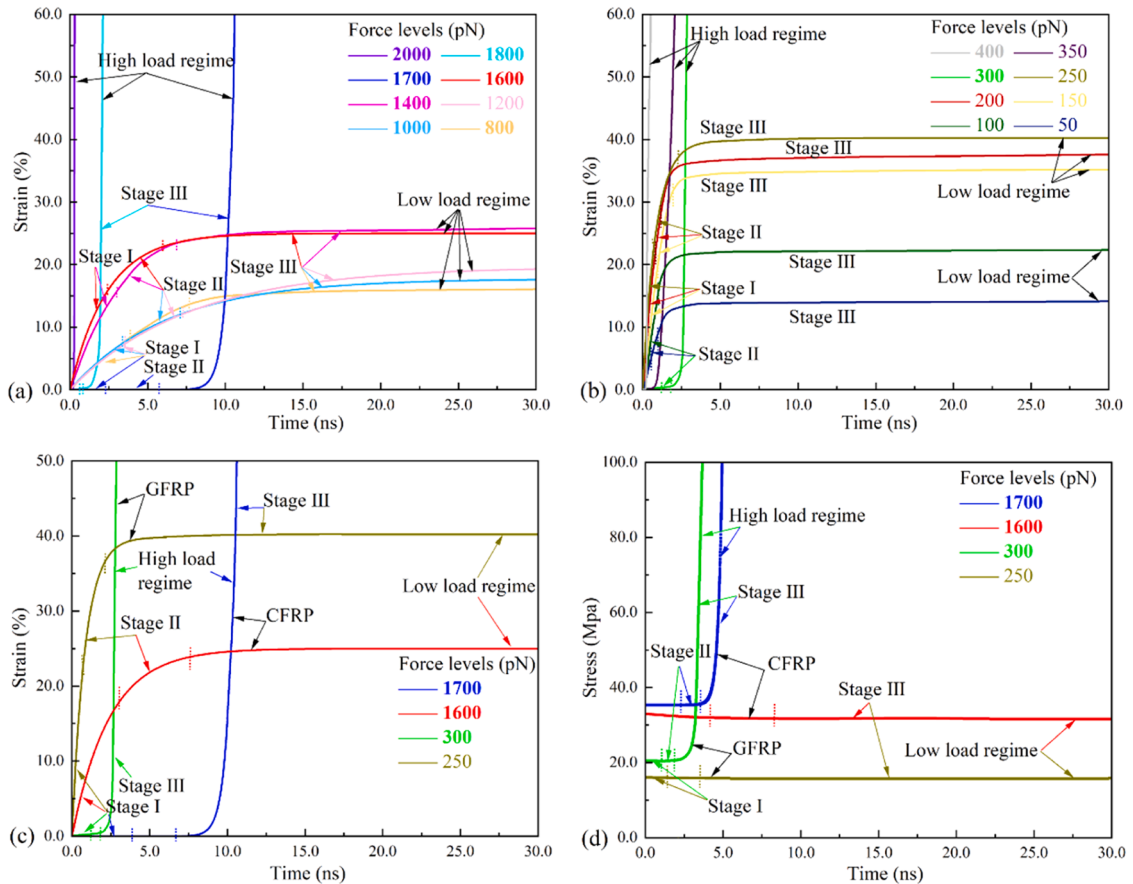


Fig. 4. Interfacial strain response for (a) CFRP and (b) GFRP; (c) comparison of strain responses at load levels of 1600 pN and 1700 pN for CFRP, 250 pN and 300 pN for GFRP; (d) comparison of stress changes at the same specified load levels as (c).

facilitate comparison of configurational alterations during the creep process, as shown in Fig. 5. It's noteworthy that the strain-time curves are divided into three stages based on the assessment of variations in the strain rate. In the first stage, strain consistently increases, as the applied load induces epoxy deformation and slight interface sliding. This results in a color shift from blue to red among specific epoxy atoms, reflecting strain-induced compression within the epoxy molecule, as depicted in Fig. 5(b) and (h). Comparing the strain curves for the carbon fiber/epoxy and glass fiber/epoxy interface in Fig. 4(a) and (b) reveals that the glass fiber/epoxy interface exhibits a steeper initial development, indicating a higher creep rate, consistent with the colored epoxy. In the second stage, the strain increase rate decelerates, along with a gradual reduction in epoxy deformation and interfacial sliding, as illustrated in Fig. 5(c) and (i). As loading continues, strain eventually stabilizes, indicating a state of equilibrium in the system's deformation, with the interface remaining intact. Under the low load regime, as depicted in Fig. 5(d), (e), (j), (k), the interface remains intact, and strain develops linearly as illustrated in Fig. 4(a) and (b). However, as the force increases into the high load regime, strain tends toward infinity, leading to interface detachment and failure, as illustrated in Fig. 5(e) and (k). Throughout the creep process, strain primarily occurs in the interfacial region adjacent to the fiber substrate, as indicated by the more distinct coloration. Additionally, it's observed that under high load conditions, the transition times between the three stages are shorter for both carbon fiber/epoxy and glass fiber/epoxy systems, as indicated by more pronounced developments in the curves.

At stress levels of 32.95 MPa and 35.01 MPa, associated with load levels 1600 pN and 1700 pN in the case of carbon fiber/epoxy, and 17.04 MPa and 20.45 MPa, associated with load levels 250 pN and 300 pN in the case of glass fiber/epoxy, these specific force levels are used to illustrate the interfacial creep evolution. Under the low load regime, the strain of glass fiber/epoxy reaches almost 40 %, surpassing the approximately 25 % observed in carbon fiber/epoxy within the same simulation time, indicating more pronounced creep. Under the high load regime, the time of detachment of the carbon fiber/epoxy interface is around 10 ns, showing an infinity curve as explained above. This detachment takes significantly longer than the 2.5 ns observed for the

glass fiber/epoxy, as illustrated in Fig. 4(c). The simulation results demonstrate that carbon fiber/epoxy exhibits less creep before reaching the threshold stress and possesses greater resistance to creep after reaching the threshold stress compared to glass fiber/epoxy. The changes in interfacial structure provide insights into the instantaneous stress variations on the fiber substrate throughout the simulation. An analysis of shear stress on the fiber substrate enhances our understanding of the interfacial response disparities between the low and high load regimes of carbon fiber/epoxy and glass fiber/epoxy. Under the low load regime, instantaneous shear stress on both the graphite substrate and glass substrate remains nearly constant, reflecting a consistent contact area during creep, as illustrated in Fig. 4(d). This observation further verifies the interface between fiber substrate and epoxy matrix remains undamaged under the low load regime. Beyond the threshold load, noticeable changes in stress occur, consistent with the strain response mentioned above. Particularly, the stress on both the graphite substrate and glass substrate remains relatively constant during the first and second stages. In the third stage, rapid acceleration of interfacial sliding leads to a significant increase in stress, approaching infinity, indicating interfacial fracture. The stress-time curves reveal that carbon fiber/epoxy can withstand higher stress levels, both at high and low stresses, and exhibits lower creep, indicating stronger resistance to creep, compared to glass fiber/epoxy. Meanwhile, macroscopic experiments have compared the creep behavior of GFRP and CFRP under different loads over 83 days [12]. The results show that as the applied stress increases, the creep process accelerates, reaching the second stage more quickly, at which point the creep rate stabilizes. Compared to GFRP, CFRP demonstrates superior creep performance, with a smoother creep rate over time curve and a longer duration before transitioning to the second stage under the same load conditions. These findings are consistent with the simulations.

3.3. Discussion of interfacial creep

It is observed that in creep simulations at varying loads, the interface transitions from attachment to detachment once the load exceeds a critical threshold, as indicated by the strength-based approach. This

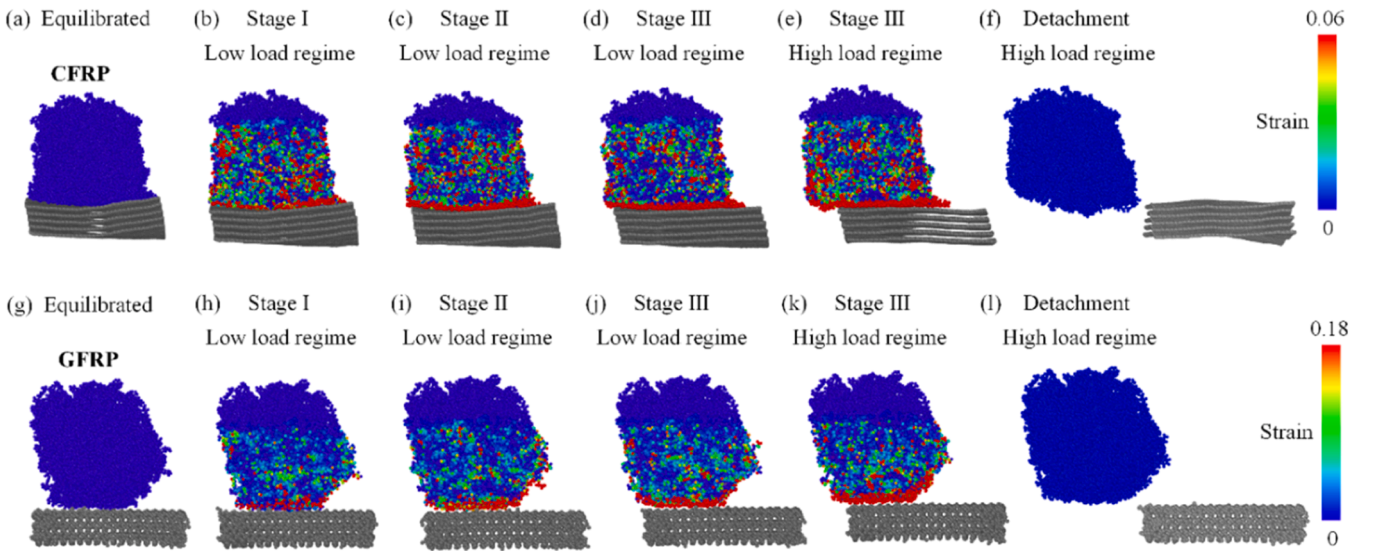


Fig. 5. The interfacial deformation under low load regime of 800, 1000, 1200, 1400 and 1600 pN (b-d), and high load regime of 1700 pN, 1800 pN and 2000 pN (e-f) for CFRP; under low load regime of 50, 100, 150, 200 and 250 pN (h-j), and high load regime of 300 pN, 350 pN and 400 pN (k-l) for GFRP; the equilibrated state for (a) CFRP and (g) GFRP. Under low load regime, the strain of the epoxy initially increases at a constant rate with slightly sliding of the interface, as shown in (b) and (h); in stage II, the increase of strain rate in the epoxy slows down while the interface continues to slide, as depicted in (c) and (i); finally, (d) and (j) illustrate stage III of the creep process, where the strain of the epoxy stabilizes and remains unchanged, and the interface remains intact. Under high load regime, the first two stages occur over a short period of time, followed by a rapid increase in strain with interfacial sliding, as shown in (e) and (k). This progression leads to the eventual detachment at the interface, depicted in (f) and (l).

transition suggests the existence of an energy barrier that marks the initiation of interfacial detachment. To enhance the comprehension of molecular-level interfacial creep behavior, Bell's analysis has been employed. This analysis correlates the transition from an attached state to a detached state at the interface with the existence of an energy barrier [33,57]. Based on Bell's model, energy barrier is quantified through the functional relationship between creep displacement of fiber substrate x_d and applied load F . This relationship is expressed by Eq. 2 [32,34]:

$$x_d = e^{\left(-\frac{E_b - Fx_b}{k_b T}\right)} x_b \omega \quad (2)$$

where E_b is the energy barrier to the initiation of interfacial detachment, x_b is the distance parameter, k_b is the Boltzmann's constant, T is the system temperature, and ω is the vibrational natural frequency. The energy barrier at the carbon fiber/matrix interface measures 15.99 kcal/mol, aligning closely with previous research that has documented values between 13.64 kcal/mol and 15.77 kcal/mol [32,56]. Similarly, the energy barrier observed at the glass fiber/matrix interface measures 2.50 kcal/mol, contrasting with the higher value of 18.69 kcal/mol reported in previous studies on debonding simulations [33]. This difference is attributed to the fact that the previous simulations involved only single-chain epoxy resins leading to a smaller value for x_b . The energy barrier represents the external energy required to initiate interfacial creep failure. This highlights that the creep behavior primarily arises from the adhesion energy between fiber substrate and epoxy matrix, driven by non-bonding interactions such as vdW and Coulombic. Once the energy barrier exceeds the interfacial adhesion energy, interfacial creep failure occurs. The calculated results reveal the adhesion energy of carbon fiber/matrix interface is higher than glass fiber/matrix interface, as depicted in Fig. 6. Additionally, to comprehensively investigate the difference in interfacial adhesion energy between carbon fiber/matrix and glass fiber/matrix interface during creep evolution, both interfaces are subjected to a loading regime of 200 pN each. Interfacial sliding at the glass fiber/matrix interface is observed immediately upon load application, indicating that the adhesion energy is exceeded under a load of 200 pN. Subsequently, interfacial sliding increased gradually with simulation time, eventually reaching a steady state. In contrast, the carbon fiber/matrix interface remains intact throughout the simulation period, as depicted in Fig. 6. This indicates that, during creep, the carbon fiber/matrix interface exhibits higher adhesion energy compared to the glass fiber/matrix interface, indicating greater resistance to creep. This

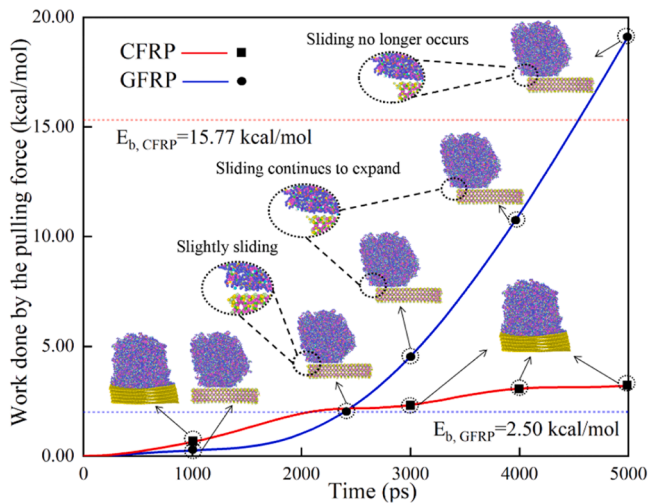


Fig. 6. The energy barrier for the initiation of interfacial detachment in CFRP and GFRP, and the work done by the pulling force of CFRP and GFRP under the load regime of 200 pN.

observation aligns with the conclusions drawn from the strength-based approach.

The energy barrier calculated through Bell's analysis precisely signifies the minimum interfacial adhesion energy required to initiate creep deformation. To gain deeper insights into energy fluctuations, particularly involving vdW and Coulombic interactions, within the interfaces of carbon fiber/matrix and glass fiber/matrix, the potential energy of bonds and non-bonded interactions in the interface region throughout the creep process under low load regime is computed. Specifically, to mitigate the influence of fluctuating contact areas A between the fiber substrate and epoxy matrix during creep, an averaging approach over the contact area A is adopted when evaluating the non-bonded energy. The outcomes are graphically represented in Fig. 7. The potential energy function is defined by the following Eq. 3 [59,60]:

$$E = E_{\text{bonded}} + E_{\text{non-bonded}} \quad (3)$$

$$= E_{\text{bond}} + E_{\text{angle}} + E_{\text{dihedral}} + E_{\text{out-of-plane}} + E_{\text{vdW}} + E_{\text{Coulombic}}$$

where E is the total potential energy, bonded interactions E_{bonded} is the bond potential energy of the system, including bond stretching E_{bond} , angle bending E_{angle} , dihedral angle torsion E_{dihedral} , improper out-of-plane $E_{\text{out-of-plane}}$, and non-bonded $E_{\text{non-bonded}}$ to describe interactions among non-bonded atoms. Fig. 7 demonstrates the consistent stability of the total potential energy of the interface models throughout the simulations, with minimal fluctuations. During creep, the bonded potential energy is primarily attributed to angle bending energy between atoms, exhibiting peak energy profile values at both carbon fiber/matrix and glass fiber/matrix interfaces, as depicted in Fig. 7(a). Notably, the non-bonded potential energy resulting from vdW and Coulombic interactions at the carbon fiber/matrix interface is significantly stable compared to that of glass fiber/matrix interface during deformation, as depicted in Fig. 7(b). Additionally, the non-bonded energy curve consistently decreases within the first two stages, indicating reduced contact area and weaker vdW and Coulombic interactions in the interface due to interfacial sliding. As the loading progresses, the energy stabilizes at a constant level, indicating an equilibrium state, consistent with the earlier mentioned creep evolution. For the carbon fiber/matrix interface, the curve shows an upward trend at stage I, primarily attributed to interfacial rebound during the creep process. This observation supports the claim that, in contrast to glass fibers, carbon fibers undergo stronger interaction forces with the epoxy resin at the contact surface, thereby contributing to an improved resistance against creep. These consistent findings are in harmony with the simulation results discussed earlier.

During creep deformation, external stress induces chain reorientation within the polymer matrix, leading to a gradual change in molecular motion over time. The creep failure process closely relates to the dynamic movement of molecular chains [60,61]. To measure the dynamics of polymer chains in the epoxy matrix under constant stress, mean square displacement (MSD) serves as a valuable metric to characterize the mobility of epoxy molecules in the system during the creep. To distinguish the effect of the fiber substrate on the upper epoxy matrix during the creep process, the epoxy resin is divided into regions of high response area and low response area based on the distance of the epoxy resin from the fiber substrate surface, as shown in Fig. 1(a). In the low response area, the motion of epoxy resin is less influenced by the forces exerted by the fiber substrate. During creep deformation, the movement of the epoxy matrix is primarily governed by vdW and Coulombic interactions with the underlying fiber substrate. The calculated MSD values at both the carbon fiber/matrix and glass fiber/matrix interfaces in the low response area are smaller than those in the high response area, as illustrated in Fig. 8. This indicates a stronger presence of non-bonding interactions experienced by the epoxy resin within the high response area. Furthermore, in both response areas, the MSD of the epoxy matrix at the glass fiber/matrix interface surpasses that at the carbon fiber/matrix interface, implying greater mobility during creep and

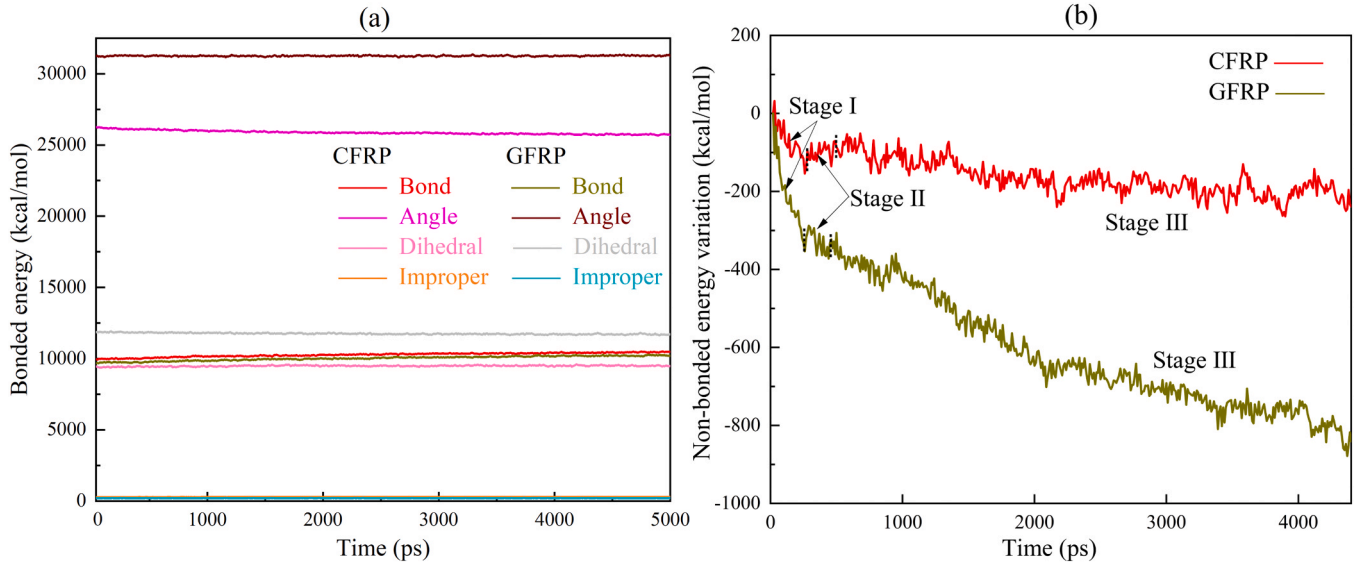


Fig. 7. (a) Bonded energy components of the CFRP and GFRP interfaces throughout the creep process, include bond, angle, dihedral, and improper energy; (b) the non-bonded energy variations of the CFRP and GFRP interfaces during the creep process, arising from vdW and Coulombic interactions between the fiber substrate and epoxy matrix.

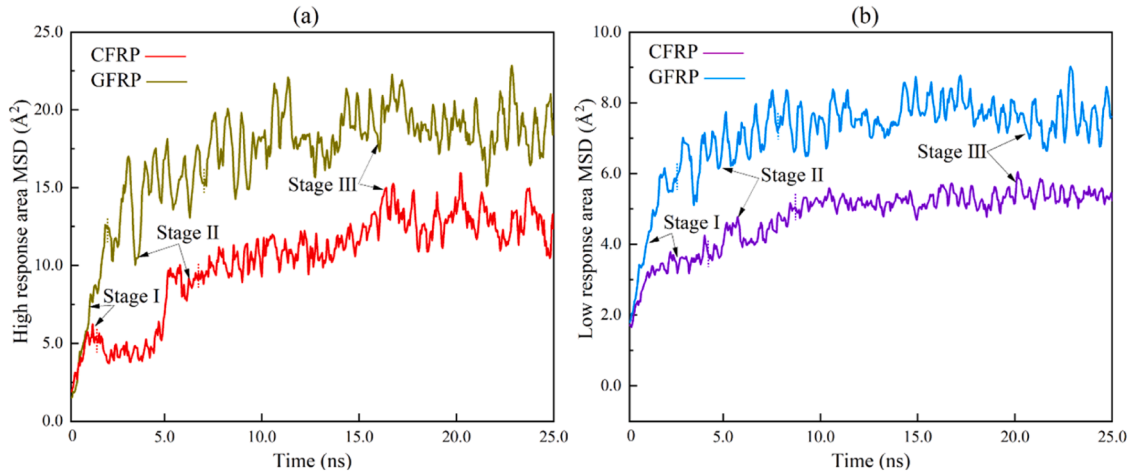


Fig. 8. The MSD of the epoxy matrix in the carbon fiber/matrix interface and glass fiber/matrix during the creep process in both (a) high response area and (b) low response area.

consequently resulting in enhanced deformation of the glass fiber/matrix interface within the same timeframe. Specifically, during the initial stage, MSD exhibits a consistent increase due to the applied load from the fiber substrate and slight sliding at the interface. Subsequently, in the second stage, the rate of MSD increase slows. As the loading process progresses, MSD eventually stabilizes, signifying that the system has reached a stable state in its deformation. This observation is consistent with the creep evolution previously described.

For further investigation and quantification of local segmental dynamics of polymer chains within the epoxy matrix, the bond autocorrelation function is frequently used, as expressed in Eq. 4:

$$C_b = \langle P_2[u(0) \cdot u(t)] \rangle \quad (4)$$

Here, P_2 denotes the second Legendre polynomial, unit vector $u(0)$ and $u(t)$ denote the direction of the bond connecting the terminal carbon atom to the central carbon atom in the epoxy matrix chain at the initial state and at time. The bracket brackets indicate the average value. This function primarily describes the changes in polymer chain mobility within composites during creep deformation. The dynamics change can

be further quantified by fitting the bond autocorrelation function to the Kohlrausch – Williams – Watts (KWW) function, as described in Eq. 5 [62]:

$$C_b(t) = C_0 e^{-\left(\frac{t}{\tau_{eff}}\right)^\beta} \quad (5)$$

where C_0 , τ_{eff} and β are fitting parameters, play crucial roles in capturing the complex relaxation behavior of polymeric systems subjected to stress. C_0 represents the initial amplitude factor, capturing rapid relaxation processes at very short timescales, such as bond and angle fluctuations associated with β relaxation. The characteristic relaxation timescale, τ_{eff} serves as a key parameter describing the timescale of relaxation phenomena. It is noteworthy that τ_{eff} is utilized as a metric to quantify changes in dynamics across different time regions, deviating from conventional structural relaxation time measurements. This approach acknowledges the coexistence of multiple relaxation processes over probed timescales [63]. Additionally, τ_{eff} indicates the time needed for polymer chains to achieve equilibrium, with a larger value signifying a more prolonged duration for generating equivalent creep deformation

in material systems. The variation in creep strain rates across various systems are attributed to the movement of polymer chains within the epoxy matrix. To explore and quantify the local segmental dynamics of polymer chains in the epoxy matrix under constant stress during creep deformation, local segmental dynamics and their corresponding relaxation times are assessed, as illustrated in Fig. 9. It is noted that mobility is markedly reduced in the carbon fiber/matrix interface, corroborating the MSD findings, as indicated by the gradual decrease in $C_b(t)$ right after the application of stress. This indicates poorer chain mobility in the carbon fiber/matrix interface, resulting in better creep resistance compared to the glass fiber/matrix interface. Moreover, trends in bond autocorrelation decay correspond to different stages of creep deformation, with rapid decay notably observed during the initial and secondary creep stages characterized by high strain rates. Fig. 9(b) and (c) present a detailed view of the microstructural evolution in the epoxy matrix, highlighting labeled polymer chains. These findings suggest that in the carbon fiber/matrix interface, the labeled polymer chains create a more intimate contact surface with the fiber substrate compared to the glass fiber/matrix interface, which induced the stretching of polymer chains at the interfacial region during the creep deformation. This enhanced proximity facilitates interaction between the polymer and substrate, leading to improved creep resistance due to the interfacial interaction between the fiber and matrix, along with the slow relaxation dynamics of the epoxy in CFRP.

To further clarify the differences in interfacial interactions between carbon fiber/matrix and glass fiber/matrix interfaces, the Connolly surface method is utilized to assess the free volume in the epoxy resin within the high response area. This method allows for the qualitative characterization of unoccupied space within the epoxy resin structures [64,65]. Blank areas on the surface indicate free volume, with larger blank areas corresponding to higher free volume values as shown in Fig. 10(a) and (b). Specifically, during the creep simulation, the epoxy resin chains migrate due to electrostatic forces. The interaction at the fiber/matrix interface primarily arises from non-bonding forces between the epoxy resin molecular chains and the atoms on the fiber surface. A higher coordination number of atoms that form non-bonding forces leads to smaller free volume and stronger interfacial interactions. Fig. 10(c) shows that the free volume concentration of epoxy resin at the CFRP interface is significantly lower than at the GFRP interface during the creep simulation. This results in polymer chains forming a more stable contact with the fiber substrate, which strengthens the polymer-fiber interaction and enhances creep resistance.

It is noticed that the fiber volume fraction, alignments, and shape significantly influence fiber/matrix interface interactions in composites, as the interfacial stress distribution and microstructural evolution of matrix will be affected by these parameters. It is reported that at fiber concentrations of 55–60 %, lower strain in the resin under sustained loading corresponds to higher interfacial interaction [66,67]. Conversely, the fatigue resistance of the fiber/matrix interface decreases significantly when fiber content exceeds 68 %. Additionally, fiber-resin

axial angles of 0° and 90° demonstrate better interfacial properties than angles of $\pm 45^\circ$. The contact area between the fiber and resin increases with higher fiber volume fraction, leading to more non-bonding sites and improved interfacial properties. However, additional fibers distribute the loads and reduce stress at the interface, it also could affect microstructure of matrix between the fibers. The geometry of fiber will further affect the specific surface area of fibers at same volume fraction, resulting in different creep behavior of composites. As these parameters are generally out of the length scale limitation of traditional full atomistic simulations at nanoscale, finite elements (FE) simulations using Representative Volume Element (RVE) can help to address these issues with the constitutive model of matrix and fiber/matrix interface provided in the presented study. Future studies on the specific effects of fiber concentration and shape on fiber/matrix interface interactions will be conducted to fill the scale gaps using the MD-FE RVE multiscale approach.

Based on the simulation results, a typical three-stage the creep deterioration process of both CFRP and GFRP composites is observed, as shown in Fig. 11. In the initial stage, stable atomic interactions form between the atoms on the fiber surface and the resin, reaching their maximum atomic coordination sites. Strain begins to develop in the epoxy resin driven by non-bonding interactions. In the second stage, slight interface sliding becomes apparent. As simulation time progresses, the strain in the resin increases, and the sliding at the interface intensifies. This leads to a gradual reduction in the atomic coordination sites involved in interatomic interaction. At this point, the non-bonding potential energy at the interface decreases, indicating a weakening of interfacial interactions. Compared to GFRP, CFRP exhibits lower resin strain and graphite provides smooth and continuous surface for the interaction with epoxy resin. Additionally, the non-bonding potential energy at the CFRP interface is higher. This suggests that the CFRP interface has stronger bonding and exhibits greater creep resistance. In high load regime, the creep process enters a detachment stage, where strain increases sharply, indicating interface detachment and damage. By examining the structural evolution of the resin at the CFRP and GFRP interfaces, the stronger creep resistance of CFRP is characterized by lower mobility and reduced free volume formation in the epoxy resin.

Macroscopic interfacial degradation can occur due to the accumulation of irreversible sliding between the epoxy resin and the fibers. Compared to the glass fiber/matrix interface, the carbon fiber/matrix interface has a smoother surface, making it more likely to recover from sliding during the same creep period, as illustrated in Fig. 11. The creep simulations reveal a clear rebound effect in the atomic dynamic trajectories, supporting this observation. This finding aligns with the interfacial bonding energy calculations mentioned earlier, indicating that the tensile force does more work at the carbon fiber/matrix interface, as shown in Fig. 6. Consequently, CFRP demonstrates higher interfacial adhesion energy and better creep resistance. The difference in creep resistance between GFRP and CFRP primarily arises from the inconsistent deformation behavior between the fibers and matrix.

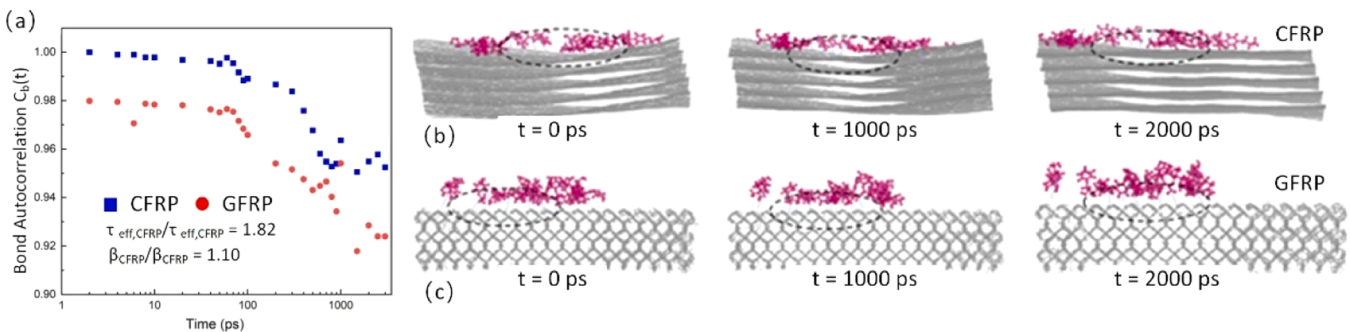


Fig. 9. (a) The bond autocorrelation decay for carbon fiber/matrix composite and glass fiber/matrix composite during the creep simulation, and the microstructural evolution in the epoxy matrix with some labeled polymer chains (b) carbon fiber/matrix composite (c) glass fiber/matrix composite.

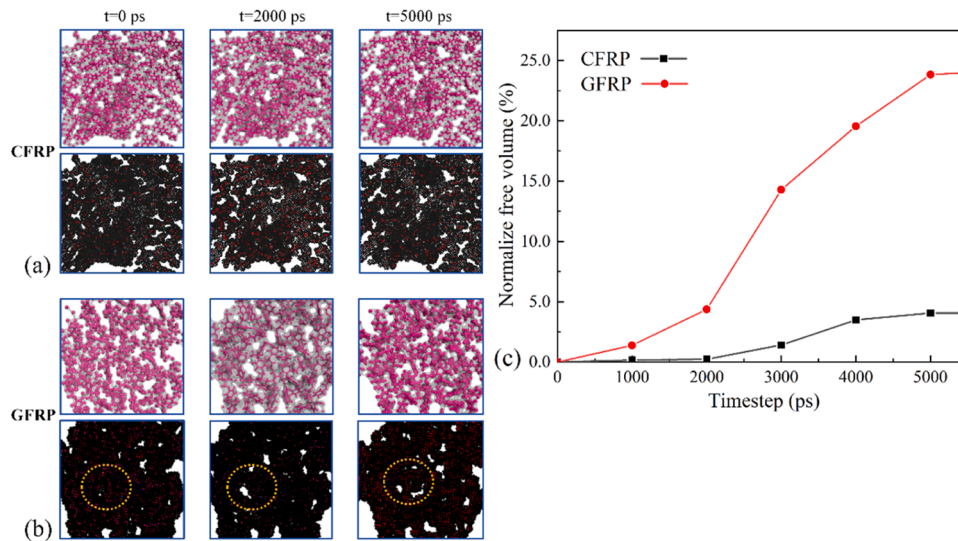


Fig. 10. The microstructural evolution in the epoxy matrix with some labeled polymer chains (a) carbon fiber/matrix composite (b) glass fiber/matrix composite; (c) normalized free volume of carbon fiber/matrix and glass fiber/matrix composite.

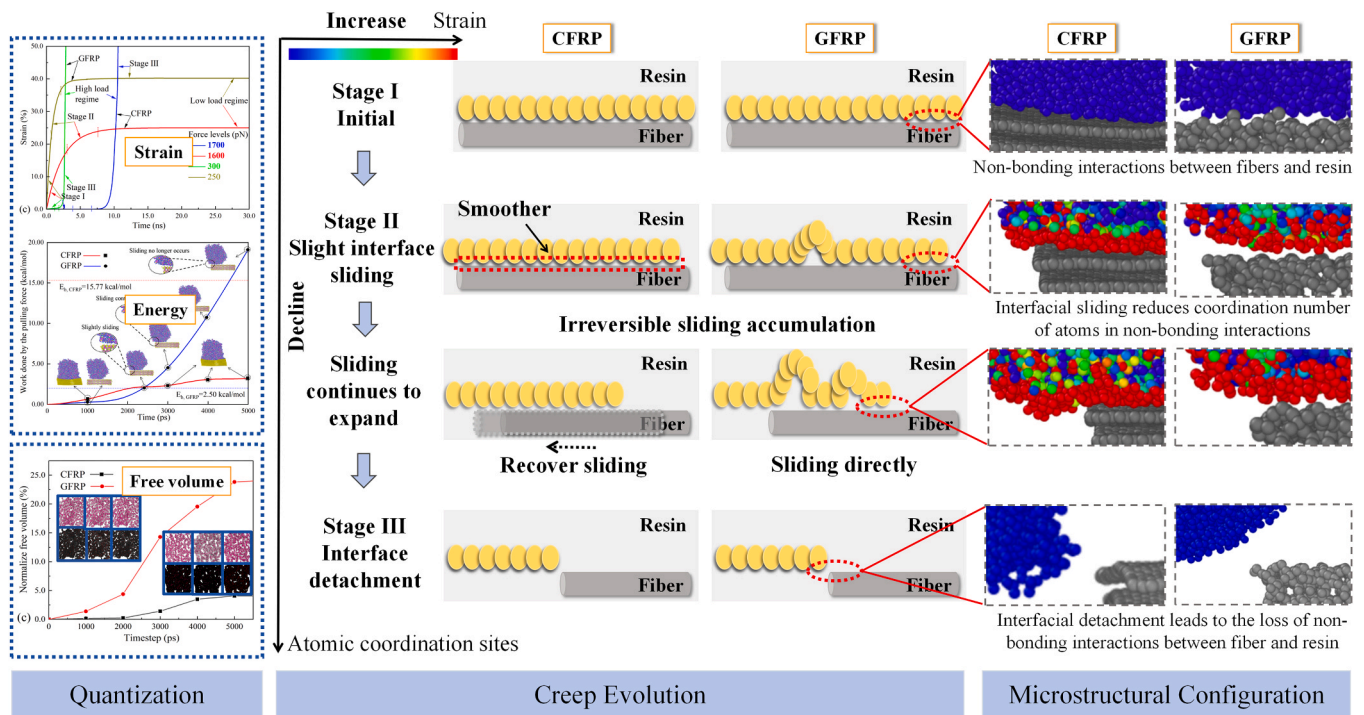


Fig. 11. Schematic diagram illustrating the microstructural evolution of creep degradation in CFRP and GFRP composites at interfaces.

4. Conclusions

In this work, SMD simulation is used to explore the creep behavior of CFRP and GFRP composites to elucidate the mechanism underlying the differences in interfacial creep behavior at the nanoscale. The findings suggest the presence of shear thresholds on the fiber/epoxy interfaces, with the threshold on the carbon fiber/epoxy interface being higher than that on the glass fiber/epoxy interface. Additionally, a comprehensive analysis of the creep evolution process at the interfaces identifies three stages of creep under low loads. Notably, under the low load regime, creep progresses through an initial stage where the interface begins to slide, followed by a gradual increase in debonding, and finally reaches a stable state where the interface remains intact. Under the high load

regime, interface sliding starts and increases rapidly, and ultimately leading to interfacial detachment. Furthermore, these results demonstrate that the carbon fiber/epoxy interface exhibits slower strain accumulation and superior creep performance compared to glass fiber/epoxy interface within the same creep duration. The energy characterization methods quantify the adhesive and non-bonding energies of the fiber/epoxy interfaces. The carbon fiber/epoxy interface exhibits higher energy values, indicating stronger interactions. Additionally, the dynamics of polymer chains in the epoxy matrix under constant stress in high response area are quantified. Glass fiber/epoxy composites show higher creep migration rates within the same creep duration, leading to more significant changes in free volume. The study findings demonstrate that both primary and secondary creep in FRP composites are influenced

by the interaction between the fiber substrate and epoxy matrix. Surface functionalization of the fiber and the use of nanomaterials are potential solutions for improving FRP composites by enhancing both bonded and non-bonded energy and slowing the movement of polymer chains at the interface.

CRedit authorship contribution statement

Ren yuan Qin: Writing – original draft, Visualization, Validation, Methodology, Investigation, Funding acquisition, Conceptualization. **Yi ze Li:** Writing – original draft, Visualization, Validation, Methodology, Investigation. **Soon Yee Wong:** Writing – review & editing, Supervision, Methodology. **Li Sun:** Writing – review & editing, Methodology, Conceptualization. **Yu Zheng:** Writing – review & editing, Supervision, Funding acquisition, Formal analysis, Conceptualization. **Fang Yenn Teo:** Writing – review & editing, Conceptualization. **Zechuan Yu:** Writing – review & editing, Visualization, Validation, Methodology, Investigation.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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